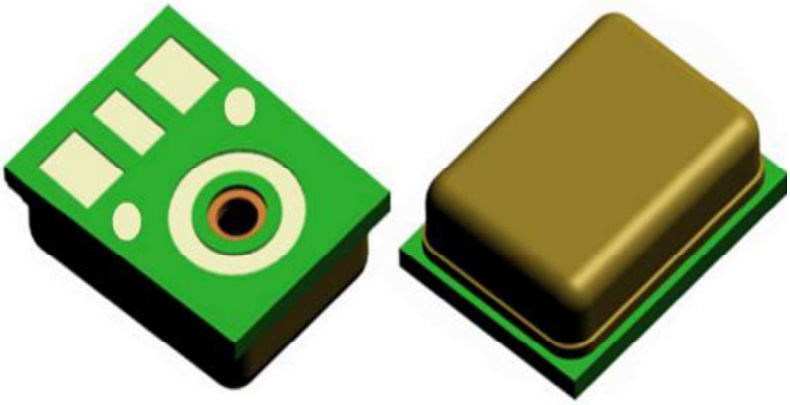
MM60-63230-B66

MEMS Microphone

The MM60 microphone series utilizes a highly robust acoustic sensor and a uniquely programmable ASIC to deliver best-in-class robustness and electroacoustic performance. This ASIC includes a low noise input buffer, output amplifier, and an integrated capacitor. The combined sensor and ASIC design provides supreme audio IMD and ultrasonic IMD with the possibility of additional ultrasonic performance enhancement by enabling a 2nd order high frequency roll off. This MM60-63230-B66 is designed for WSA F1/T0 project.

Product Features

- Trimmable : Sensitivity, Phase and Current
- Tight acoustic tolerance ±1 dB
- Omnidirectional
- Internal Cap : 2.2 nF
- Packaging : SPA
- Port Location : 12Jn
- Acoustic port dimension : Ø 0.30 mm
- Overall height : 1.30 mm



RoHS COMPLIANT

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Acoustic and Electrical Specifications

Test Condition	
Temperature	20 ± 5 °C
Humidity	50 ± 25% R.H
Supply Voltage	0.9 V, unless otherwise stated
Electrical Load	No load, unless otherwise stated
Sound Source	Pressure Cavity

General Specifications	Min	Nom	Max	Units	Condition
Polarity	Increasing output voltage				Relative to negative terminal with increasing inlet pressure
*Weight	--	0.0212	--	g	

Electrical Specifications	Min	Nom	Max	Units	Condition
Supply Voltage	0.88	0.90	1.40	V	Operating range including supply noise and transients
Output Load Voltage (V_L)	0.40	0.50	0.65	V	after settled (open load)
Output Impedance	1200	2700	4200	ohms	--
Power Supply Rejection Ratio (PSRR)	38.0	45.0	-	dB	75 mVpp sine wave @ 1 kHz
Supply Current (Drain)	35	40	45	µA	--
*Startup Time @ 94dB SPL	-	-	9.0	s	Sensitivity stable within 0.23 ± 0.15dB Both tolerance extremes don't happen in same reel

Acoustic Specifications		Min	Nom	Max	Units	Condition
Sensitivity	100 Hz	-2.0	-1.0	0.0	dBV	dB re sensitivity @ 1kHz
	1 kHz	-36.5	-35.5	-34.5		dB re 1V / 1 Pa
	10 kHz	-3.0	-2.0	-1.0		dB re sensitivity @ 1kHz
*Resonance Peak	Amplitude	-	7.5	-		dB re sensitivity @ 1kHz
	Frequency	-	19.0	-	kHz	-
*Low Frequency Roll-Off (LFRO)		-	50.0	-	Hz	-3 dB re sensitivity @ 1kHz
Input-Referred Noise, A-Weighted		-	25.0	26.5	dB SPL	100 Hz to 20 kHz bandwidth
Signal to Noise Ratio, A-Weighted		67.5	69.0	-	dB A	100 Hz to 20 kHz bandwidth
*Input-Referred Noise, 1/3rd Octave	100 Hz	-	12.0	-	dB SPL	-
	1 kHz	-	9.9	-		-
	10 kHz	-	15.5	-		-
*Acoustic Overload Point (AOP)		-	119	-	dB SPL	10% THD @ 1 kHz
*Input SPL amplitude @ distortion		-	118	-	dB SPL	5% THD @ 1 kHz

Other Specifications		Min	Nom	Max	Units	Condition	
*Input-Referred EMI Noise	0.7-0.96 GHz	-	7.0	-	dB SPL	@ 90 V/m	According to IEC 60118-13 Ed. 4.0
	1.4-2.0 GHz	-	1.0	-	dB SPL	@ 50 V/m	
	2.0-2.7 GHz	-	1.0	-	dB SPL	@ 35 V/m	
*Input Referred Vibration	Z axis	-	65	-	dB SPL/g	1 g acceleration (Open Port), @ 1 kHz	
	Y axis	-	51	-			
	X axis	-	< 30	-			
*Humidity Coefficient		-	0	-	dBV	per %RH	
*Storage Temperature		-40	-	105	ºC	-	
*Operating Temperature		-17	22	63	ºC	-	
*Electro-Static Discharge Tolerances		MIL – STD – 750 Class 2 Rating ; EOS / ESD – S5.1 – 1993 Class 2 Rating					

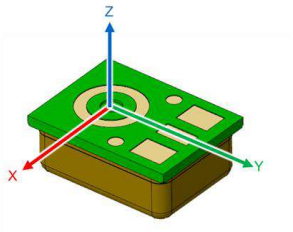
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Acoustic and Electrical Specifications

Matching Specifications		Max Deviation	Units	Condition
Phase	250 Hz	| 1.5 |	degrees	Microphone with greater phase response value at 250 Hz is marked with blue stripe
Sensitivity	1k Hz	| 1.5 |	dB	
Phase	250 Hz	| 7.0 |	degrees	Front microphone of all pairs
	1k Hz	| 4.0 |		

Notes

- * Denotes parameter not tested in production, values are measured during product development.
- Performance may vary in conditions beyond the specified 'Test Conditions'.
- Exposure beyond those indicated under 'Acoustic & Electrical Specifications' for extended periods may affect device reliability.
- Vibration orientations shown: -

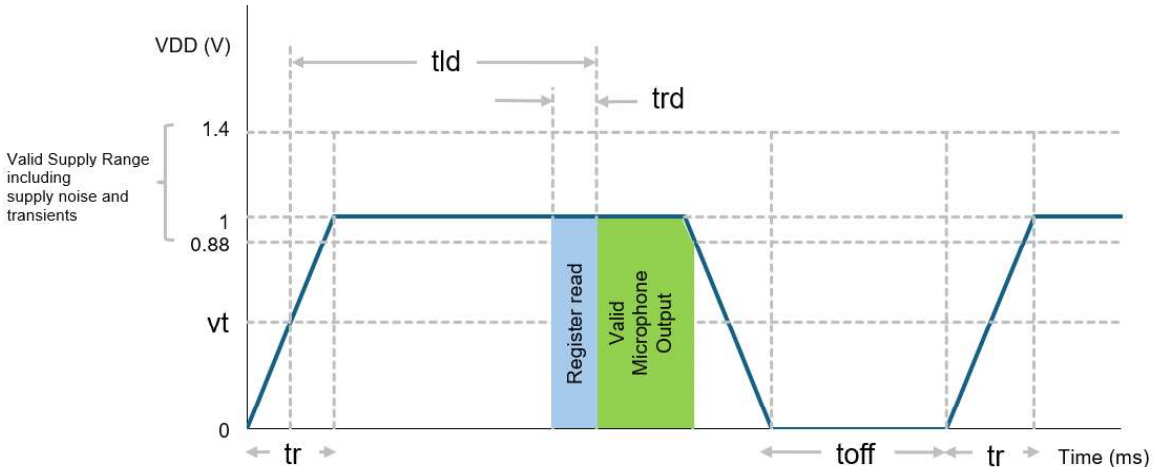


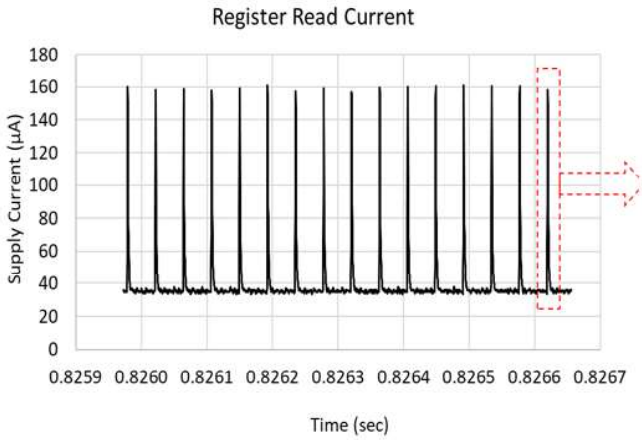
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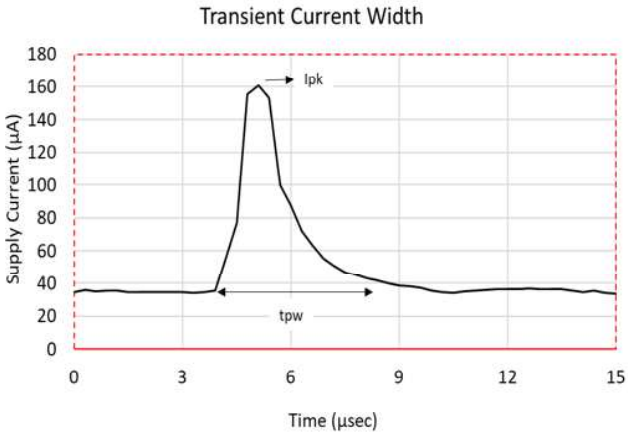
Start-Up Specifications

Test Condition	
Temperature	-17 to +63 °C

Specifications	Symbol	Min	Nom	Max	Units	Condition
*Supply Ramp time	tr	--	--	125.0	ms	Time to reach minimum supply voltage
*Supply off time	toff	250.0	--	--	ms	Supply off between power cycles
*Register loading time	tld	--	--	630.0	ms	Time to complete register loading
*Supply threshold	vt	0.30	0.50	0.75	V	Minimum threshold level for startup trigger
*Startup Peak Battery Drain	Ipk	--	150.0	200.0	uA	Peak transient current
*Transient current width	tpw	--	--	5.0	µs	Pulsing current duration
*Register read duration	trd	--	220.0	--	µs	Register read duration





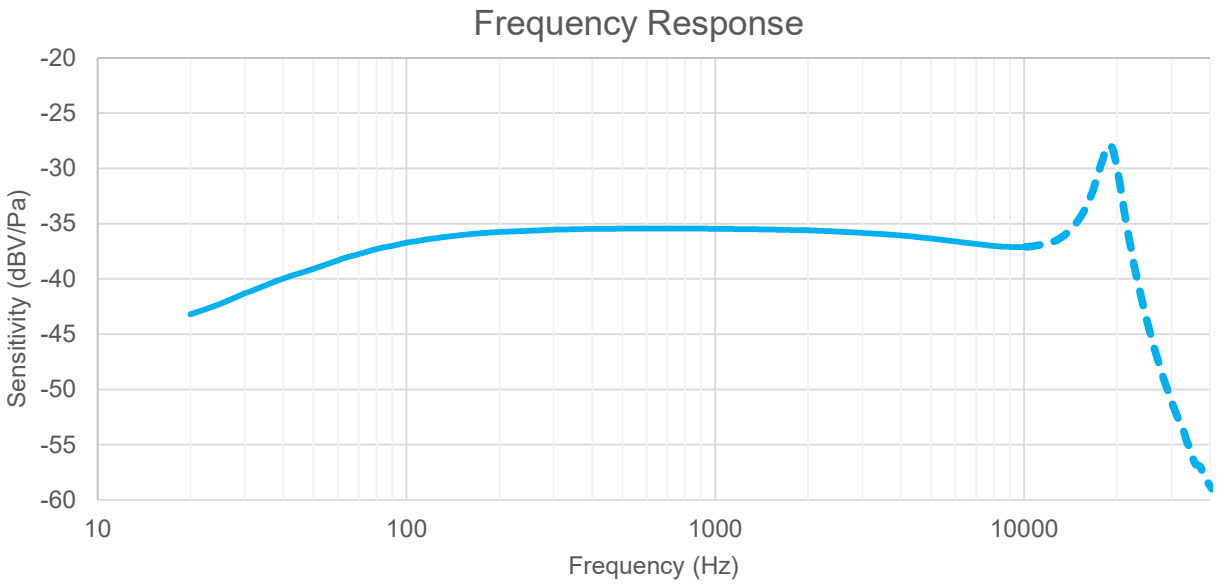


Notes

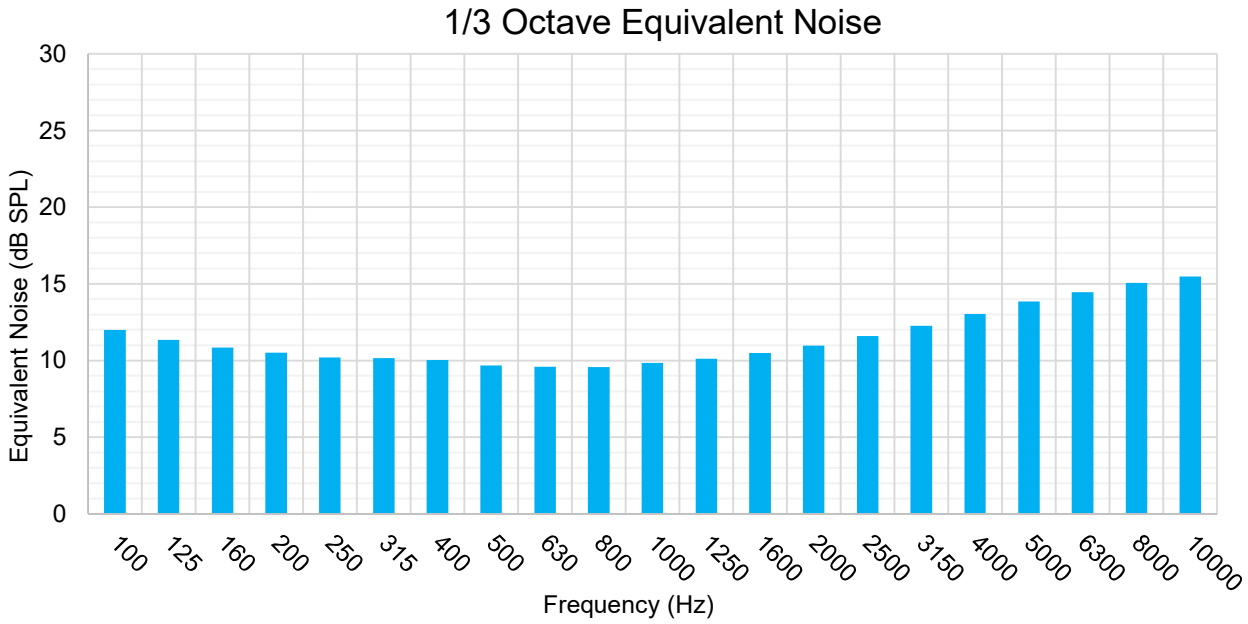
- * denotes parameter not tested in production, values are measured during product development.
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Typical Performance Curves



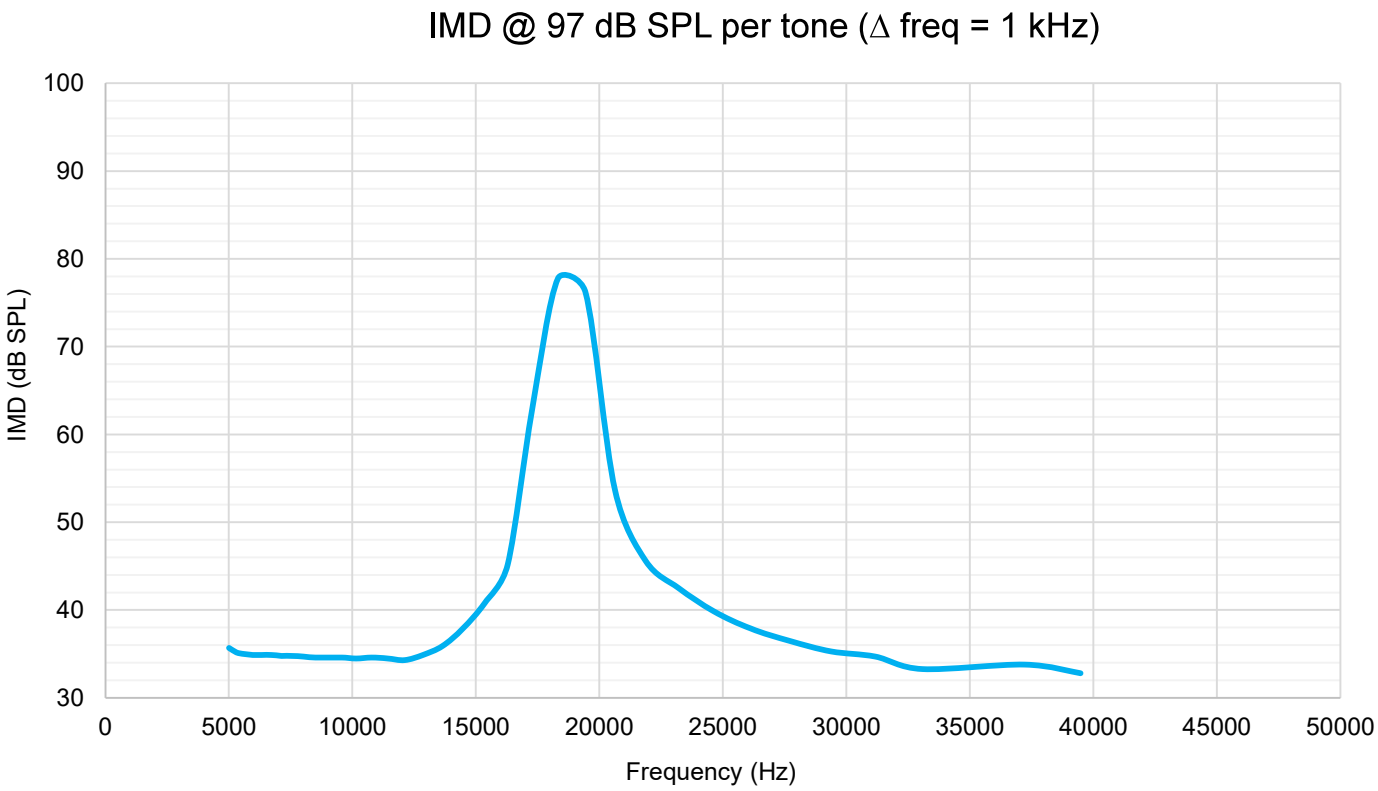
- Typical frequency response measured in a pressure cavity under nominal conditions.
- Response shown in dotted line is not tested in production, values are measured during product development

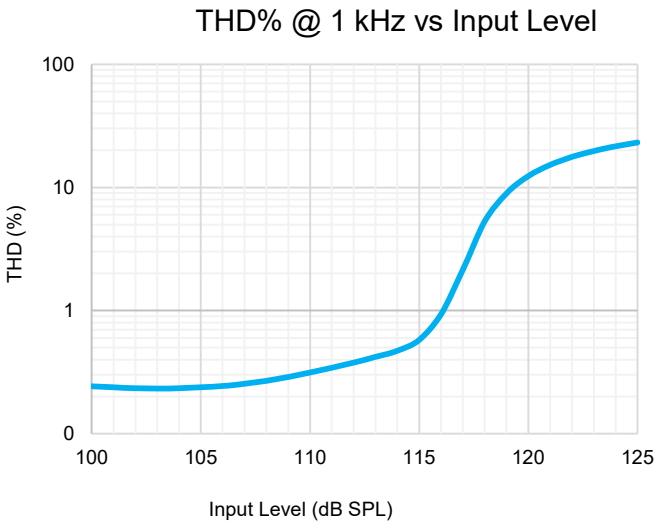


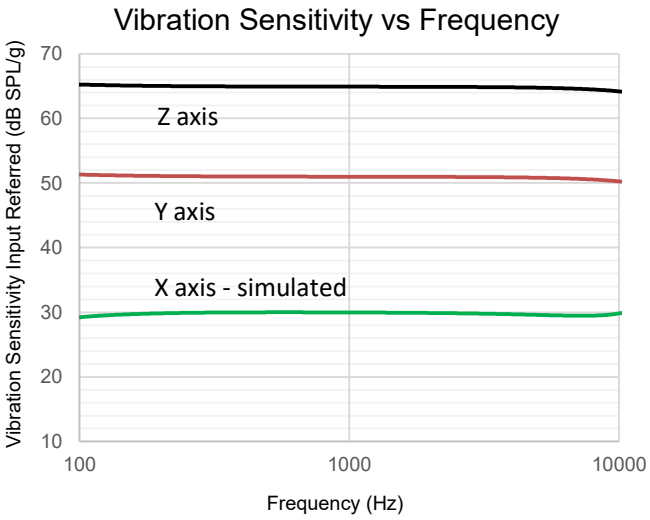
- Typical Equivalent Noise under nominal conditions

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Typical Performance Curves



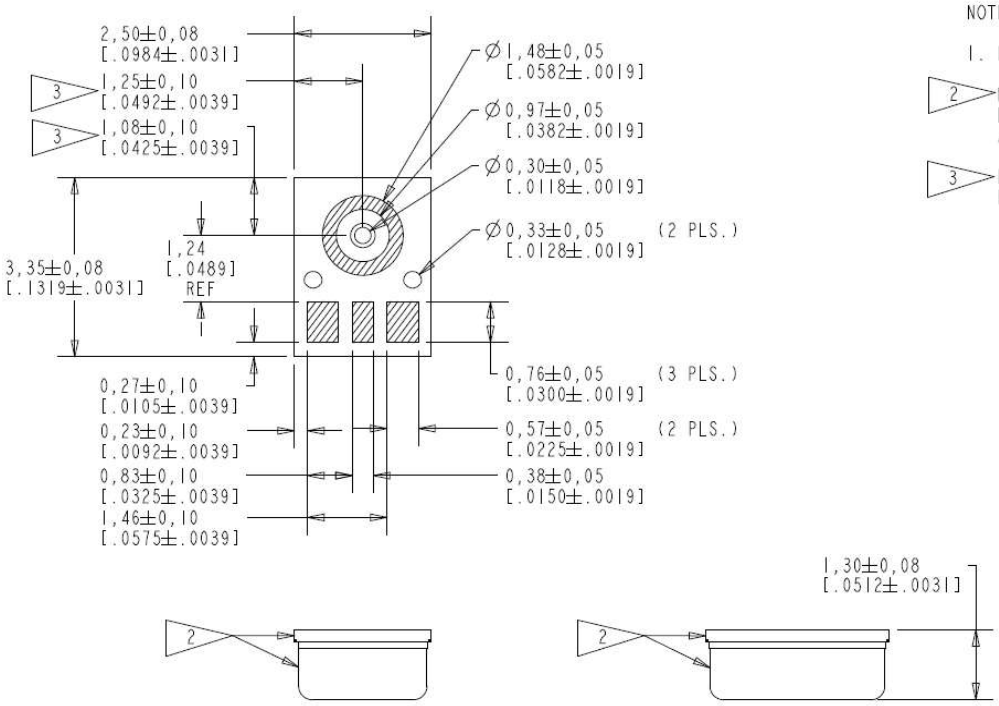




- The simulated X axis curve is shown as the measurement is below the test system noise floor.

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Mechanical Specifications



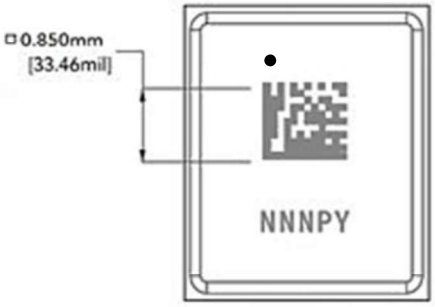
NOTES:

1. DIMENSION IN MILLIMETERS [INCHES].

2. METAL CAN MAY BE FLUSH WITH, BUT NOT EXTEND PAST, EDGE OF MICROPHONE BASE ON ALL FOUR SIDES.

3. DIMENSION APPLIES TO THE CENTER OF GROUND RING.

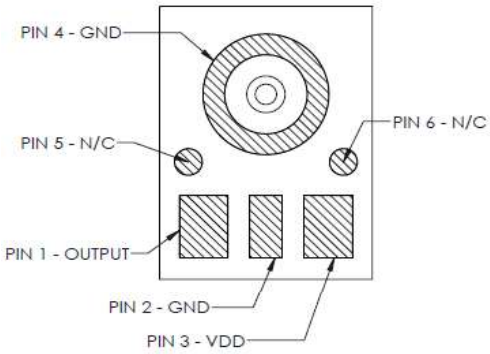
Symbol	Definition	# Characters	Value
N	Reference Number	3	180
P	Production Week	2	01 to 52
Y	Production Year	1	0 to 9



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Mechanical Specifications

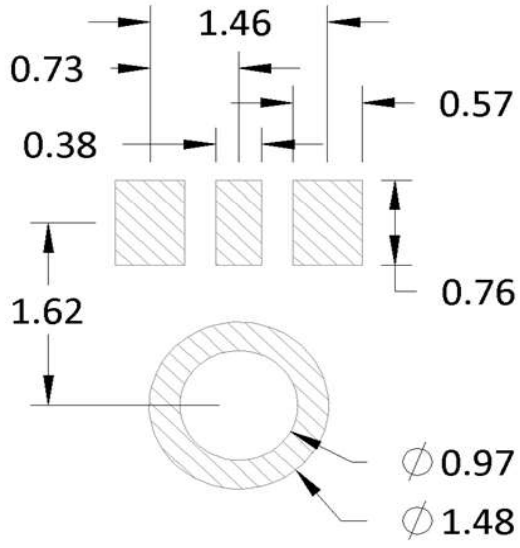
Pin #	Name	Type	Description
Pin 1	OUT	Signal	Output Signal
Pin 2	GND	Power	Ground
Pin 3	VDD	Power	Power Supply
Pin 4	GND	Power	Ground
Pin 5	N/C	N/A	Test Pins (Knowles Internal Use)
Pin 6	N/C	N/A	Test Pins (Knowles Internal Use)



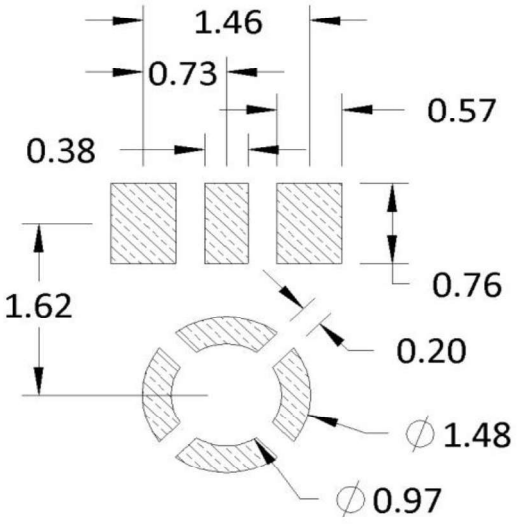
BOTTOM VIEW
DETAIL OF PIN-OUT
(HATCHED AREA INDICATES
SOLDERABLE SURFACE)

Notes
VDD and OUT must not be connected to either Pin 5 or 6.

Example Land Pattern



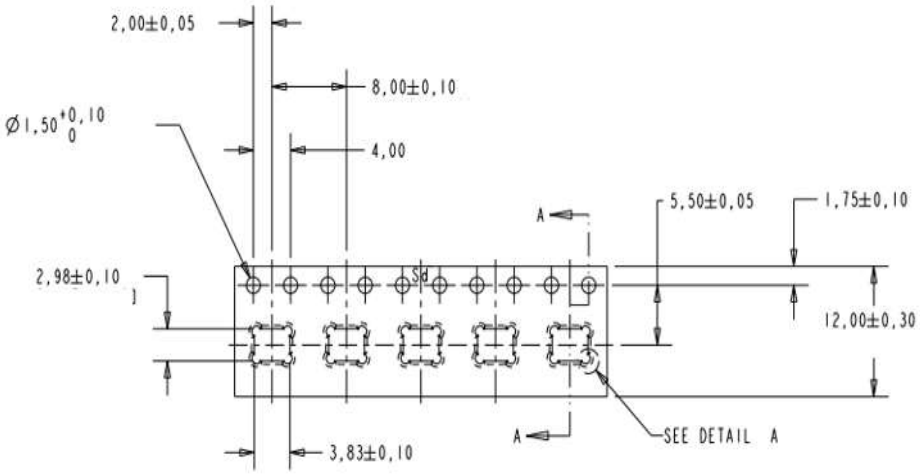
Example Solder Stencil Pattern

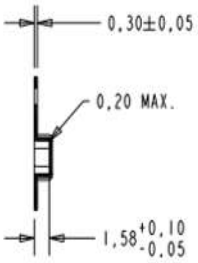


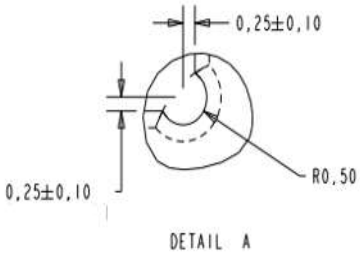
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Packaging

Parameter	Details	Units
Reel Diameter	7	inch
Quantity per reel	500	pairs
Empty pockets at the start and end	60	pcs

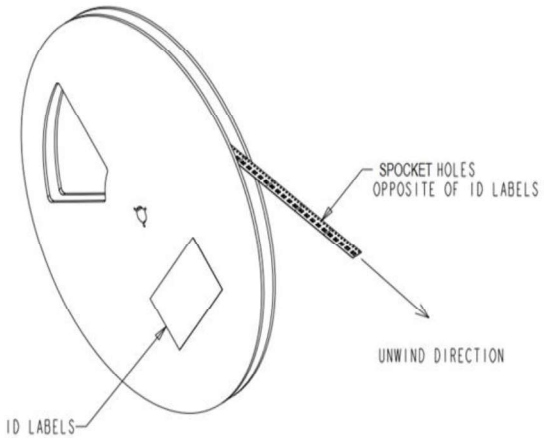






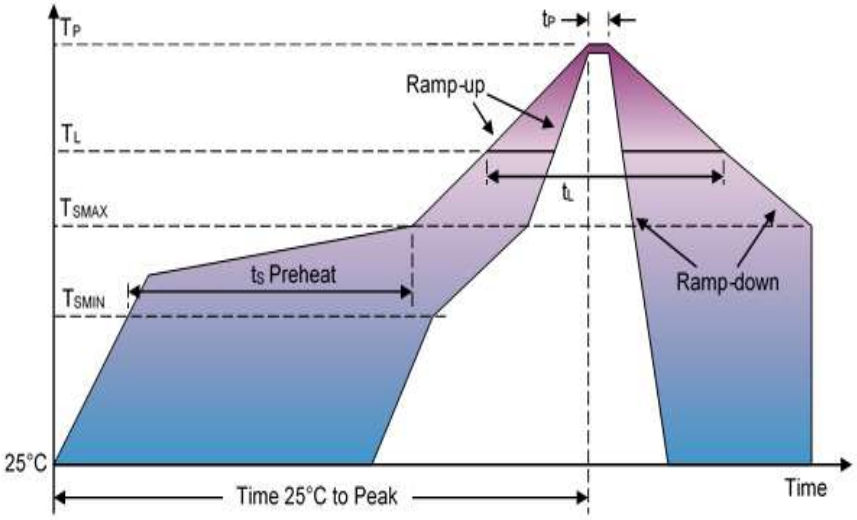
Notes:

- Dimensions are in millimeters unless otherwise specified.
- Units packed in sequence Lead Mic, Lag Mic, Empty Pocket, Lead Mic ...
- Tape & reel per EIA-481.
- Labels applied directly to reel and external package.
- Shelf life: Twelve (12) months when devices are to be stored in factory supplied, unopened ESD moisture sensitive bag under maximum environmental conditions of 30 °C.



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Recommended Reflow Profile



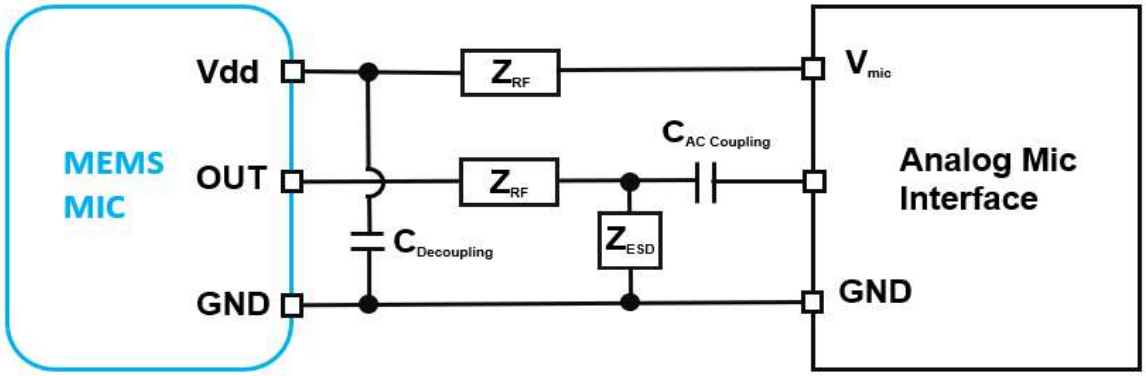
Profile Feature	Pb-Free
Ramp-up rate (25 °C to T_{SMIN})	3 °C/second (see note)
Average Ramp-up rate (T_{SMAX} to T_P)	3 °C/second max
Preheat •Temperature Min (TSMIN) •Temperature Max (TSMAX) •Time (TSMIN to TSMAX) (tS)	 150 °C 200 °C 60-180 seconds
Time maintained above: •Temperature (TL) •Time (tL)	 217 °C 60-150 seconds
Peak Temperature (T_P)	260 °C
Time within 5 °C of actual Peak Temperature (t_P)	20-40 seconds
Ramp-down rate (T_P to T_{SMAX})	6 °C/second max
Time 25 °C to Peak Temperature	8 minutes max

Notes

- Based on IPC/JEDEC J-STD-020 Revision C.
- All temperatures refer to topside of the package, measured on the package body surface.
- Ramp-up rate (25 °C to TSMIN) can be optimized based on application.
- MSL (moisture sensitivity level) Class 1.
- Maximum of 3 reflow cycles is recommended.
- In order to minimize device damage:
  - a) Do not expose microphone to liquid or particles, especially at the tube and vent hole.
  - b) Do not board wash or clean after the reflow process.
  - c) Do not brush board with or without solvents after the reflow process.
  - d) Do not directly expose to ultrasonic processing, welding, or cleaning.
  - e) Do not insert any object in port hole of device at any time.
  - f) Do not apply over 120 psi of air pressure into the port hole.
  - g) Do not pull a vacuum over port hole of the microphone.
  - h) Do not apply a vacuum when repacking into sealed bags at a rate faster than 0.5 atm/sec.
  - i) Do not directly expose to vapor phase soldering.

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Recommended Application Circuit



Notes

- All Ground pins must be connected to ground.
- RF and ESD performance can be improved by placing optional components close to the microphone pads: -
  Z_{RF} - inductive components, such as ferrite beads.
  Z_{ESD} - Zener diode
- Decoupling capacitors should be place next to Vdd pin of the microphone for best performance; 1uF is recommended.
- Refer to AN-13 for soldering guideline; available at www.Knowles.com.

Materials Statement

- Meets the requirements of the European RoHS directive 2011/65/EC as amended.
- This product meets RoHS and REACH requirements as of date of publication. To receive latest certificate of compliance, send email to environmental@knowles.com

Reliability Tests

Test	Description
Reflow	5 reflow cycles with peak temperature of +260 °C
High Temperature Storage	+105 °C environment for 1,008 hours (JESD22-A103E.01)
Low Temperature Storage	-40 °C environment for 1,008 hours (JESD22-A119A)
High Temperature Bias	+105 °C environment while under bias for 1,008 hours (JESD22-108G)
Low Temperature Bias	-40 °C environment while under bias for 1,008 hours (JESD22-108G)
Temperature/Humidity Bias	+85 °C / 85% R.H. environment while under bias for 1,008 hours (JESD22-A101D.01)
Thermal Shock	100 cycles of air-air thermal shock from -40 °C to +125 °C with 15 minute soaks (IEC 68-2-14)
Vibration - Sine	16 minutes in each X, Y, Z axis from 20 to 2,000 Hz with peak acceleration of 20 G (MIL 883L, Method 2007.3,A)
Mechanical Shock	3 pulses of 10,000 G in each of the X, Y, and Z directions (IEC 68-2-27 Test Ea)
ESD-HMM: air-to-porthole discharge	10 Discharges at ±6V, ±8V, ±10V, ±12kV, ±16kV. with 150pF, 330Ω air discharge lowering towards porthole until discharge.
ESD-HBM	3 discharges of ± 2 kV direct contact to I/O pins (ANSI/ESDA/JEDEC JS-001-2023)

Specification Revisions

Revision	Specification Changes	Date
A	Introduced	Apr 16 2026

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